

Web Development Fundamentals

Start your web journey with HTML, CSS, and JavaScript. Build real projects from day one.

Welcome to the course! This document contains all the materials, lessons, and exercises you will need to successfully complete the curriculum. We recommend reading through this guide carefully and taking notes as you progress.

Category: Development | Level: Beginner | Est. Time: 6 Hours

Course Objectives & Overview

By the end of this course, you will have a solid understanding of the core concepts, practical applications, and best practices associated with this topic. This structured material is designed to guide you step-by-step from foundational theories to advanced problem-solving techniques.

Ensure you complete all practical exercises and review the lesson summaries before attempting the final exam.

Lesson 1: Introduction to HTML

HTML (HyperText Markup Language) is the foundation of every web page. In this lesson you will learn:

- What HTML is and why it matters
- The basic structure of an HTML document: , , ,
- Common HTML tags: headings (h1-h6), paragraphs (p), links (a), images (img)
- Lists: ordered (ol) and unordered (ul)
- Forms and inputs: , ,
- Semantic HTML: , , , ,
- HTML attributes: id, class, href, src, alt
- Nesting elements correctly
- Comments in HTML

Key concept: HTML describes the structure and content of a web page. It is NOT a programming language — it is a markup language.

Key Takeaways:

- Review the main concepts outlined above.
- Practice the techniques discussed.
- Write down any questions for further research.

Lesson 2: CSS Styling Basics

CSS (Cascading Style Sheets) controls the visual presentation of your HTML.

- Selectors: element, class (.), id (#), attribute
- The box model: margin, border, padding, content
- Colors: hex, rgb, hsl, named colors
- Typography: font-family, font-size, font-weight, line-height
- Flexbox: display:flex, justify-content, align-items, gap
- CSS Grid: grid-template-columns, grid-template-rows
- Responsive design with media queries
- CSS Variables: --my-color: #ff0000
- Transitions and basic animations
- Pseudo-classes: :hover, :focus, :nth-child

Key concept: CSS cascades — later rules override earlier ones. Specificity determines which rule wins.

Key Takeaways:

- Review the main concepts outlined above.
- Practice the techniques discussed.
- Write down any questions for further research.

Lesson 3: JavaScript Basics

JavaScript adds interactivity and logic to your web pages.

- Variables: var, let, const
- Data types: string, number, boolean, null, undefined, object, array
- Operators: arithmetic (+, -, *, /), comparison (===, !==, <, >), logical (&&, ||, !)
- Control flow: if/else, switch, ternary operator
- Loops: for, while, forEach, for...of
- Functions: declaration, expression, arrow functions
- DOM manipulation: querySelector, getElementById, addEventListener
- Events: click, input, submit, keydown
- Fetch API for HTTP requests
- Async/Await and Promises

Key concept: JavaScript runs in the browser and can modify the page dynamically without reloading.

Key Takeaways:

- Review the main concepts outlined above.
- Practice the techniques discussed.
- Write down any questions for further research.

Conclusion & Next Steps

Congratulations on reaching the end of the written material for this course! Your next step is to log into the platform and take the final assessment. Review your personal notes, ensure you understand the key takeaways from each lesson, and apply what you have learned.

Good luck!